

LECTURE

Infix, Postfix, Prefix exp

$0, \dots, 9, -, +, *, /, (,) \{$

$(2)^* (4+3)^* (2+6) /$

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Infix, Postfix, Prefix exp.

$A+B$ $3/2$ $5+3$ 2^4

$A+B \Rightarrow$ Postfix

$A \oplus B$ convert $\boxed{A-B+}$
 $A \oplus B$ Post $\Rightarrow AB-$

Infix, Postfix, Prefix exp.

$$5+3, 2^4$$

Prefix $\rightarrow$ $A+B$

$A+B$

$+AB$

Convert $\rightarrow$ postfix, Prefix

$$① \quad \underbrace{(A+B)}_{①} \underbrace{C}_{②}^*$$

Post $\rightarrow$

$$(AB+)C^*$$

Scan $\downarrow$

$$(A+B)C^*$$

$$[A+B]C^*$$

Post

$$② \quad \text{Prefix} \quad (+AB)^*C$$

$$② \quad * (+AB)C$$

$$③ \quad A+BC^*$$

DATA STRUCTURES

② $(A^B)^*C$

① $(AB^*)^*C$
 ② $(AB^*)^*C$
 ③ AB^*C^* Postfix

① $(^AB)^*C$
 ② $*(^AB)C$
 ③ $*^AB C$ Prefix

② $(A \div B)^*C$

③ $A * B^C$ → $A * (B^C)$
 (ABC^{^*})

Convert → postfix, Prefix

① $(A \oplus B)^*C$

Post → ①
 (AB+) C

(A = C) *
 (H = C) *

① $(^AB)^*C$
 ② $*(^AB)C$
 ③ $*^AB C$

② Prefix $(+AB)^*C$
 ③ $*^+AB C$

pro-
 $A * (^BC)$

$*A^BC$

DATA STRUCTURE

(2) $A \wedge B \wedge C \wedge D$

$A \wedge B \wedge (C \wedge D)$ ① $A \wedge (B \wedge (C \wedge D))$
 $A \wedge (B \wedge (C \wedge D))$ ② $A \wedge (B \wedge (C \wedge D))$
 $A \wedge (B \wedge (C \wedge D))$ ③ $ABCD$

$A \wedge (B \wedge C \wedge D)$

$A \wedge B \wedge C \wedge D$

① $A \wedge B \wedge (C \wedge D)$

② $A \wedge B \wedge (C \wedge D)$

③ $A \wedge (B \wedge (C \wedge D))$

$ABCD$

Convert $\rightarrow$ postfix, prefix

(1) $(A \oplus B) * C$

post $\rightarrow$ ① $(AB+)$ *

$(AB+)$ *

$(AB+)$ *

① $(A \oplus B) * C$
 ② $(+AB) * C$
 ③ $* (+AB) C$

③ $A + A B C$